ILIA + EXPO-RT

Small action edits for household robots

Can learned corrections help complete daily tasks?

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Challenge submission / October 2026



Actual rollout / task 002

Six everyday tasks

Human demonstrations show the behaviors we want the robot to learn.



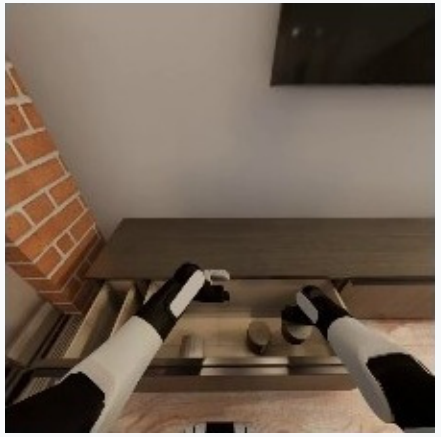
000

Radio



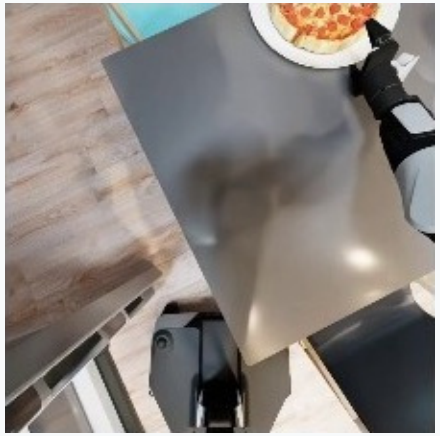
001

Trash



002

Decorations



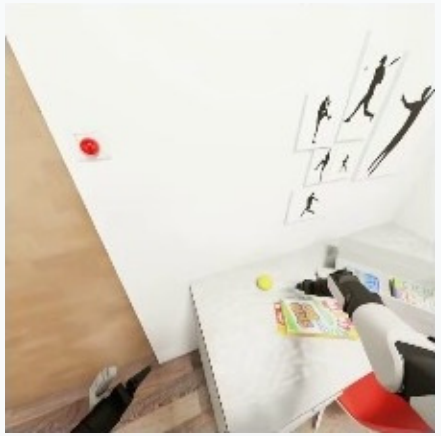
003

Plates and food



005

Mousetraps



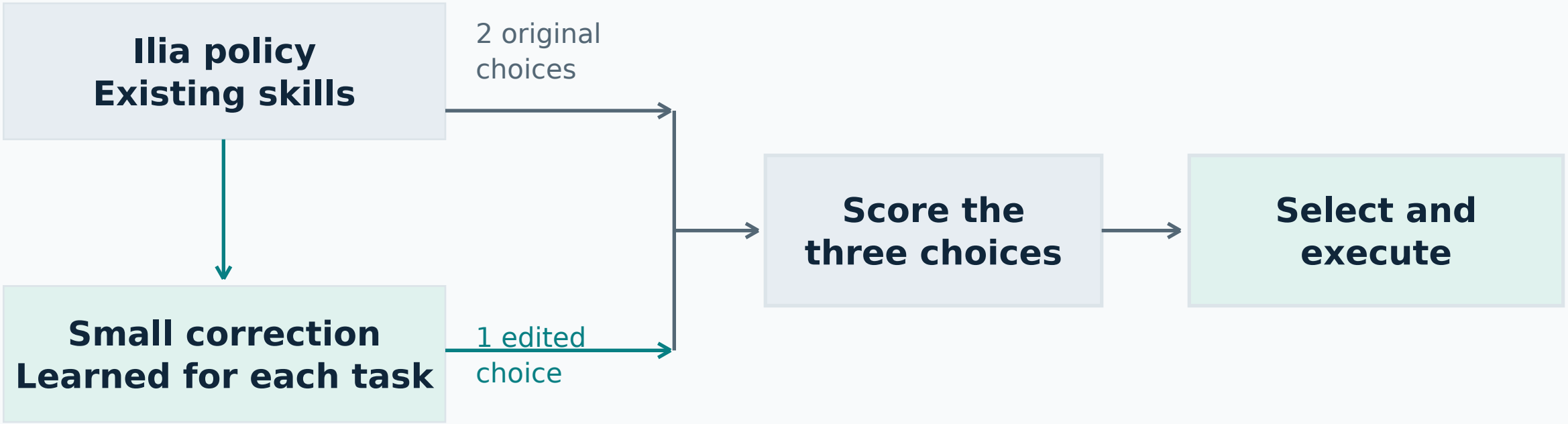
007

Toys

Source: original human demonstration panels from the previous presentation.

Ilia proposes actions. EXPO-RT edits them.

The original robot policy stays frozen.



Ilia keeps its stage tracking and recovery rules.

The added networks guide action choice. Predicted value is separate from task success.

Training, comparison, then submission

The development comparison and final evaluation answer different questions.

20 demonstrations per task → **1,000** training updates

DEVELOPMENT PILOT

36 rollouts

6 tasks, 3 instances

Original policy versus RT

FINAL EVALUATION

12 rollouts

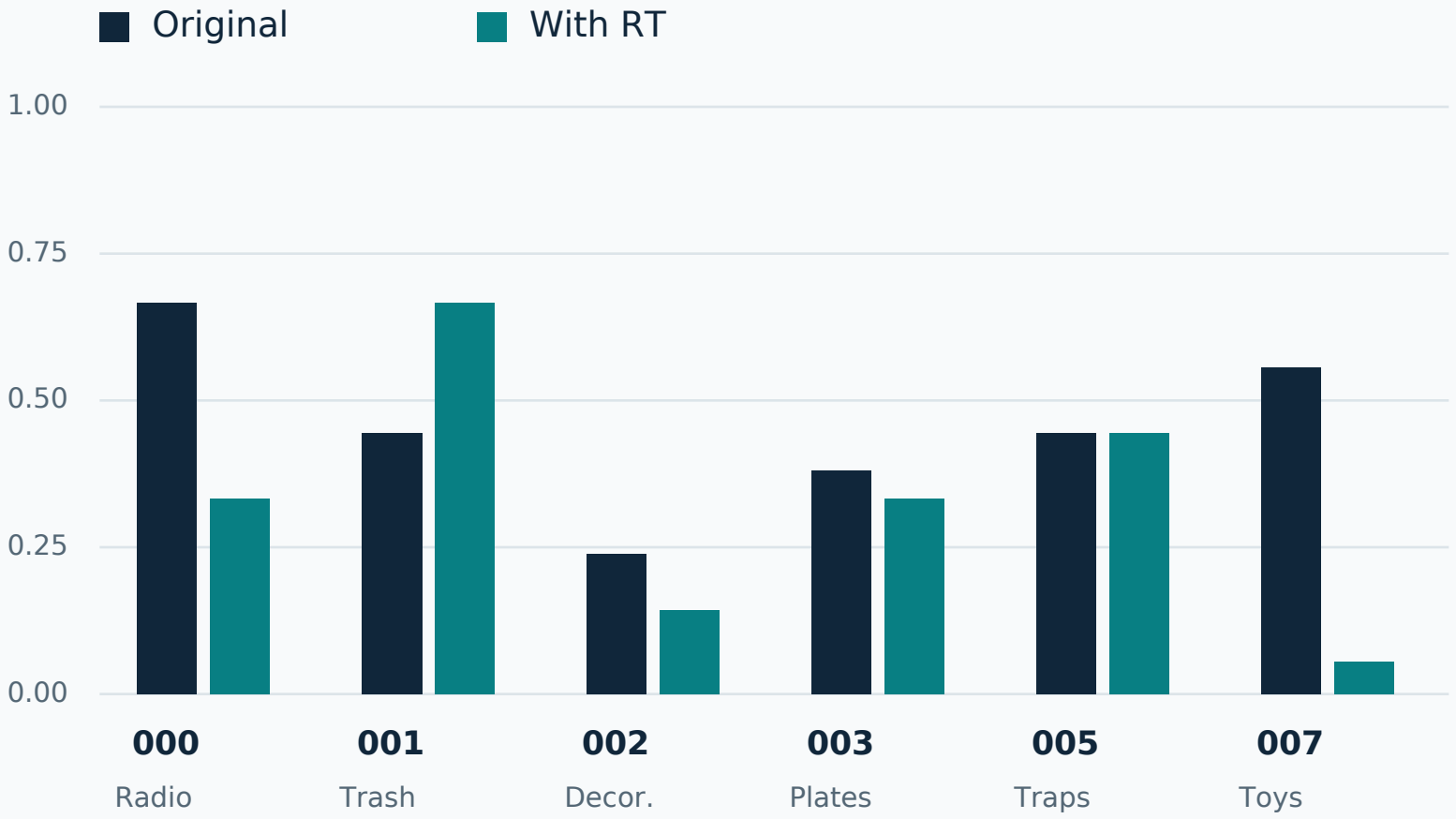
3 tasks, public instances 0-3

RT only, one run per instance

The final evaluation has no matched original-policy comparison.

RT did not improve the pilot overall

Mean final goal progress across three development instances per task.



MEAN Q

0.455

0.329 with RT

COMPLETE SUCCESSES

5/18

1/18 with RT

Trash gained partial-goal progress, while full-task successes fell.

Partial progress, no complete successes

All 12 final runs are included. Values show goals satisfied at episode end.

	Public 0	Public 1	Public 2	Public 3
002 Decorations	57%	71%	57%	0%
001 Trash	0%	0%	0%	0%
005 Mousetraps	0%	50%	0%	33%

0/12

complete successes

22.4%

mean goal progress

Q measures progress. Complete success requires all task goals.

Self-reported. With 988 unsubmitted instances counted as zero, full-challenge Q = 0.00269048.

Trash: handling objects did not finish the task

Task 001 / public instance 0



60 s

Approach



132 s

Hold the bin



237 s

Repeated holding pose

The goal: place three cans inside the bin.

Likely bottleneck: insertion and recovery. Video samples do not prove the exact cause.

Simulation time. Head view on the right, wrist views on the left. Final Q: 0.0 in all four instances.

Decorations: outcomes varied by instance

The same policy reached different levels of final goal progress.



Public 0 **57% final Q**

Further object interactions



Public 3 **0% final Q**

Close objects block wrist views

Frames at 500 s and 230 s. Occlusion and stalling are observations, not confirmed causes.

Mousetraps: some goals remained unfinished

Final recorded views from the two runs with partial goal progress.



Public 1 **50% final Q**



Public 3 **33% final Q**

No complete successes in four runs. Final JSON determines scores; images show the task context.

The submission is complete

The portal accepted the entry on October 11, 2026.

01

Code

Policy and evaluation instructions

02

Evidence

12 original videos and result files

03

Docker

Embedded model weights

Organizer verification is pending.

L40S deployment checks passed. Actual 24GB hardware remains untested.

Next research priorities

Planned work, separate from the submitted policy.

01 Placement

Measure whether the object reaches and stays in the target region.

02 Recovery

Detect repeated poses and learn how to release, retreat, and retry.

03 Fair comparison

Compare original and edited actions on the same instances and settings.



An existing rollout, not a new experiment.

Explore the evidence

Public files preserve the experiment and support reproduction.

Project overview	Method, results, and current status
Evaluation guide	Docker launch and exact evaluation setup
Original rollout videos	All 12 recordings, including every failure
Research paper	Technical details, analysis, and data appendix

Data: frozen pilot and final evaluation records. Images: actual demonstrations and rollout frames.

Illustrative frame selections explain behavior; the original videos and metric files stay unchanged.